Adaptive Trust-Aware Retrieval-Augmented Generation for Medical Evidence: Guarding Against Prompt Injection and Retrieval-Induced Hallucination

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***Abstract*—Retrieval-augmented generation (RAG) lets a large language model (LLM) draw on external documents, but retrieval does not make evidence reliable. A medical retriever can return passages that are irrelevant, outdated, attached to the wrong drug, or written to manipulate the model, and a conventional pipeline treats all of them alike once they reach the context window. We built a trust-aware RAG prototype for drug-drug interaction (DDI) and adverse drug event (ADE) questions that holds every retrieved item as untrusted data. Before generation, a chunk is excluded if its provenance check fails, its content hash does not match, it carries instructions addressed to the model, it names none of the requested medications, or its source identity is unverified; the remaining chunks must reach a risk-tiered trust threshold. After generation, each claim is checked by a pinned natural-language-inference (NLI) model at clause level, by a drug-pair identity check, by citation checks and by a medication identity gate, and a claim is released only when all of them agree. We report an offline engineering evaluation. The frozen candidate reproduced byte for byte, with 2,252 backend tests passing in normal and network-blocked modes. On a constructed, author-known set of 113 statements, the identity gate passed none of 54 unsupported statements (95% upper bound 6.6%) and wrongly withheld 2 of 59 valid ones; across seven development sets no unsupported statement was released, and recomputing the release decision from the recorded checks showed that without the identity gate the other checks would have released 3 to 19 identity-unsupported statements per set. 509 scoped security tests passed. Two defects found by tracing withheld statements (a one-way vocabulary rule and a guard that read ordinary words as brand names) were fixed in a remediation candidate, which changed 6 of 731 decisions with no regression. In a small pilot with two hosted LLMs on 30 synthetic medication sets, neither model had an answer released for any of the 10 control sets whose medication pair has no evidence in the corpus, and 1 and 2 answers were released overall; a vision model read 35 of 36 names on the 14 of 20 synthetic prescriptions it answered. The labels are AI-written and unreviewed, the main evaluation used scripted generation, and no comparison with a conventional semantic RAG baseline has been run, so we make no claim about clinical accuracy or about reducing hallucination relative to such a baseline. The prototype is for research and decision support, not for diagnosis or prescribing.**

***Index Terms*—adverse drug events, drug-drug interactions, evidence verification, large language models, prompt injection, retrieval-augmented generation, retrieval poisoning, trust-aware retrieval**

I. Introduction

A. Background and motivation

LLMs are now capable at biomedical question answering and clinical text processing, and their use on medication questions has grown quickly. Their knowledge sits in model parameters, and an answer produced from parameters alone carries no record of where each statement came from. RAG addresses part of this gap: the original formulation paired a parametric generator with a non-parametric document store [1], the approach has grown into a broad family of methods [2], later work added adaptive retrieval and self-critique [3], and medical benchmarks have examined how retrieval choices affect medical question answering [4].

Medicine adds a requirement that general RAG does not address: a passage that is semantically close to the query is not, for that reason, trustworthy. Medication safety shows why. Radha Krishnan et al. compared ChatGPT with pharmacist assessments of DDIs for 120 hospitalised patients and found minimal agreement (Cohen's κ 0.077 to 0.143) [5]. Sicard et al. found that ChatGPT, Claude and Gemini detected most DDI cases in pharmacovigilance reports but had low specificity and reported non-existent interactions for negative controls [6]. Both studies point to the same need: a medication answer is only as safe as the evidence that can be checked behind it.

B. Problem statement

A conventional medical RAG pipeline is

$$Q \;\rightarrow\; \mathcal{R}(Q) \;\rightarrow\; \mathcal{C} \;\rightarrow\; \mathcal{G}(\mathcal{C}, Q), \tag{1}$$

where Q is the query, $\mathcal{R}(Q)$ the retrieved evidence, $\mathcal{C}$ the context assembled from it and $\mathcal{G}$ the generation step. Nothing in (1) tests whether $\mathcal{C}$ is reliable: it may hold irrelevant or outdated evidence, evidence about the wrong medication, low-authority sources, instruction-bearing text or poisoned content. We use the following terms. *Prompt injection* is text intended to manipulate model behaviour; *indirect* injection arrives through retrieved content [7]. *Retrieval poisoning* manipulates the corpus so that misleading content is retrieved

[8]. A *retrieval error* is unsuitable evidence retrieved without an attacker. *Retrieval-induced hallucination* is unsupported or incorrect generated content to which unsuitable retrieved evidence contributed. The research problem is how a medical RAG system can assess the trustworthiness and security of retrieved evidence before that evidence influences the answer, and verify the answer afterwards.

C. Research questions

1) Can evidence-trust assessment make medical retrieval more reliable than semantic relevance alone?
2) Can prompt-injection and retrieval-poisoning controls reduce the influence of malicious retrieved content?
3) Can identity and cross-source checks identify pharmacological claims that are misattributed or weakly supported?
4) Can claim and citation verification reduce unsupported generated statements?
5) Can controlled abstention improve safety when evidence is insufficient or contradictory?
6) What latency do the evidence controls add?

This paper reports evidence on questions 2 to 6 at the level of software behaviour on constructed data (for question 3, its identity side); question 1 and the clinical side of question 3 need data that does not yet exist (Section X).

D. Contributions

The individual components (trust scoring, adversarial filtering, retrieval, claim verification) are not new on their own. Our contribution is combining them in one implemented pipeline and evaluating it reproducibly:

1) an evidence-eligibility stage that excludes chunks by provenance, content hash, embedded instructions, medication alignment and verified source identity before a nine-factor, risk-tiered trust score is applied;
2) a release rule under which a claim is shown only when its support state, a medication identity gate and a medication guard agree, and no later stage can restore a rejected claim;
3) an evaluation method that reports where each statement stopped (before the gate, at the gate, after it, released) instead of a single accuracy figure, with exact or Wilson intervals, and recomputes the release decision under subsets of the checks from recorded results to isolate what each check adds;
4) a reproducible offline evaluation (frozen datasets, hashes, run records) and a remediation candidate whose two fixes changed 6 of 731 decisions with no regression and no unsupported release;
5) a small pilot of the same release path on output from two hosted LLMs, and of prescription-image extraction with a vision-language model.

II. Related Work

LLMs and medication safety. LLM answers on drug interactions depended on prompt wording and agreed only minimally with pharmacists [5], and across three LLMs specificity was low, with interactions reported for negative controls [6]. Hakim et al. called for guardrails in pharmacovigilance, such as detecting anomalous documents and incorrect drug names before generation [9]. RAG-based decision support for prescribing errors across 16 specialties worked best with a pharmacist in the loop [10], and in medication-safety education grounded in drug package inserts GPT-4o scored 0.86 against 0.89 for clinical experts [11].

Medical RAG, hallucination and verification. Medical RAG applies RAG [1] to biomedical text [4], where terminology must be normalised [16] and sources differ in authority. Hallucination in generation is well documented [12], and RAGTruth shows that unsupported and contradictory claims persist with retrieval [13]. FActScore checks atomic facts against a source [14]; Self-RAG lets the model critique its own retrieval and output [3]. We instead compute trust and security outside the model. Retrieval uses BM25 [19], dense embeddings [17], [18] and reciprocal rank fusion [20]; verification uses NLI [21] with a biomedical model [22], [23].

Prompt injection and poisoning. Indirect prompt injection compromises real LLM-integrated applications [7], OWASP lists prompt injection among the main LLM application risks [15], and PoisonedRAG showed that a few crafted texts can steer RAG output [8].

III. Threat Model

The adversary can influence the user input, external documents or the retrieval corpus. Each threat has an implemented control: direct prompt injection meets query screening and sanitisation; indirect injection in a chunk excludes the whole chunk; retrieval poisoning meets a provenance check, a poisoning-score filter and verified source identity; evidence tampering is caught by comparing the SHA-256 of the text with its registered hash; entity mismatch meets the medication alignment filter, the identity gate and the drug-pair identity check; contradictory evidence is handled by clause-level NLI stance and an aggregate check; and citation mismatch by resolving each marker to an eligible chunk and checking it. Natural contradictions in the literature are not treated as attacks by default.

IV. Implemented Architecture

Fig. 1 shows the components of the final candidate (C2j), a Python backend that streams its stages to a browser client; the development results were measured on C2h and C2i, whose release path C2j keeps unchanged. The design treats retrieved information as untrusted until it has passed relevance, trust, security and verification controls. Two simplifications relative to a full production design matter for reading the results: the evaluated corpus is local (adapters for PubMed, Europe PMC and openFDA exist and are unit-tested but are not wired to the live path), and adversarial screening is rule based.

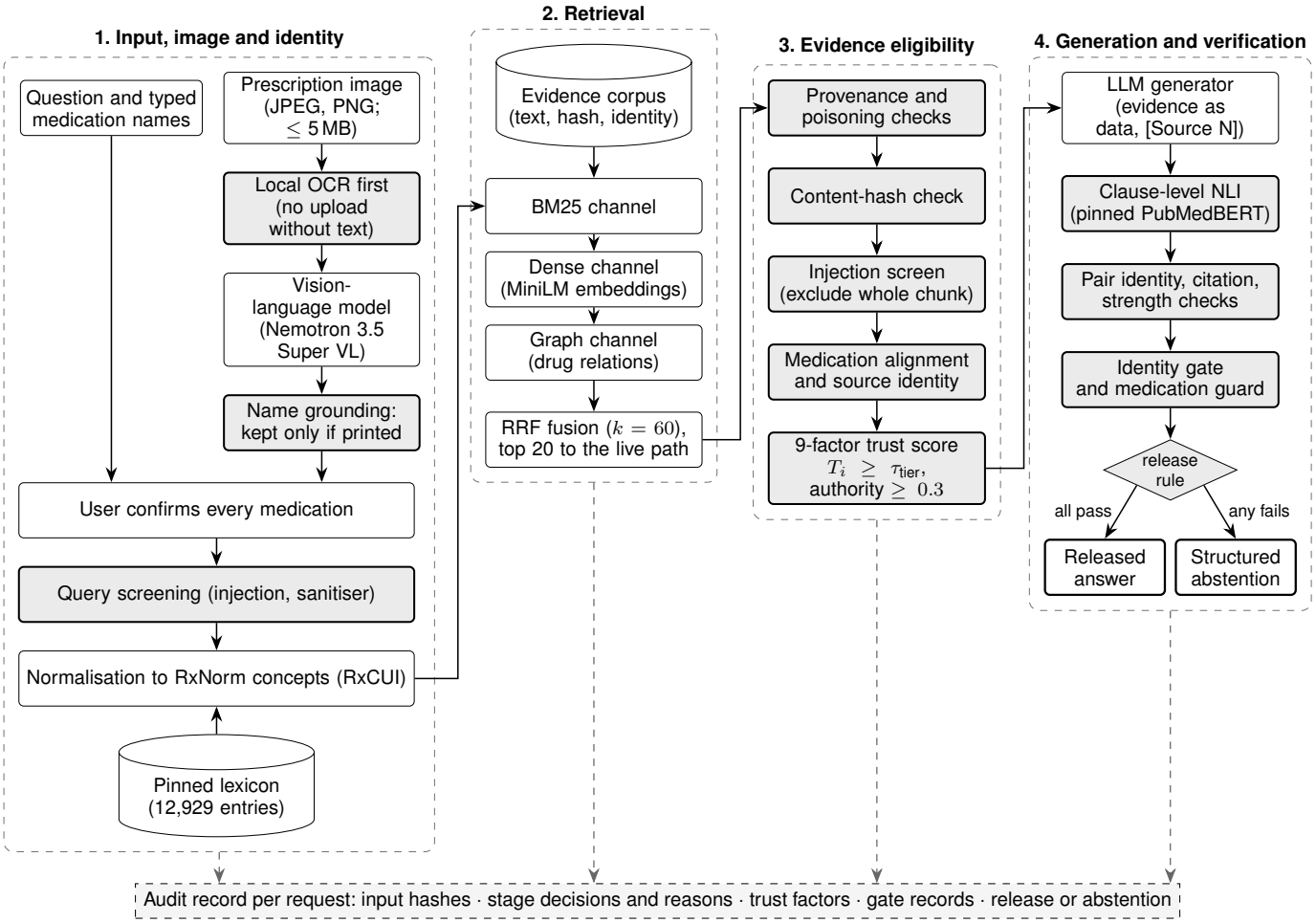


Fig. 1. Architecture of the final candidate (C2j). Shaded nodes are the controls added to a conventional RAG pipeline. An uploaded image is read by local OCR before anything leaves the machine; the vision model's names are kept only if they are printed on the image, and the user confirms every medication. A failed exclusion check removes a chunk; a failed claim check withholds the whole answer.

A. Input, identity and retrieval

Medication names are extracted from the request, confirmed by the user and resolved against a pinned RxNorm prescribable lexicon [16] (12,929 entries) and a curated RxCUI cache; brands, ingredients and salts remain distinct concepts. A prescription image is first read by local OCR; it is sent to the vision-language model only if that OCR finds text, and a returned name is offered for confirmation only if it is printed on the image. Retrieval runs three channels over the same corpus: BM25 over lower-cased tokens [19], cosine similarity over `all-MiniLM-L6-v2` embeddings [18], and an adjacency-list graph of drug relations seeded with the confirmed medications. Reciprocal rank fusion

$$\mathrm{RRF}(d) = \sum_{r \in \mathcal{L}} \frac{1}{k + \mathrm{rank}_r(d)}, \qquad k = 60, \tag{2}$$

merges the lists [20]; at most 50 fused candidates are kept, chunks with a poisoning score above 0.4 are dropped, and the live path requests the top 20.

B. Evidence eligibility and trust

Each candidate e_i passes five exclusion checks: a provenance poisoning check; equality of the SHA-256 of its text with the registered content hash; an injection screen that excludes the whole chunk (stripping the instruction would change the hashed evidence); medication alignment (the chunk must name a requested medication); and verified source identity. Duplicate texts are kept once. Surviving chunks receive a trust score over nine factors $f_{ik} \in [0, 1]$,

$$T_i = \sum_{k=1}^{9} w_k^{(r)} f_{ik}, \qquad \sum_k w_k^{(r)} = 1, \tag{3}$$

whose weights $w^{(r)}$ depend on the risk tier $r \in \{\mathrm{R0}, \ldots, \mathrm{R3}\}$ of the question (Table I). A factor that the evidence record does not carry is treated as missing and contributes zero, so a gap never raises the score. The eligibility decision is

$$G_i = \big[\,\text{passes all exclusions}\,\big] \wedge \big[\,T_i \geq \tau_r\,\big] \wedge \big[\,A_i \geq 0.3\,\big], \tag{4}$$

with $\tau = (0.30, 0.45, 0.60, 0.75)$ for R0 to R3 and A_i the source authority. Adversarial risk enters as hard exclusions

TABLE I
TRUST FACTOR WEIGHTS PER RISK TIER (CONFIGURATION OF C2H)

Factor	R0	R1	R2	R3
Source authority	0.15	0.20	0.25	0.30
Query relevance	0.20	0.20	0.15	0.10
Evidence quality	0.10	0.15	0.20	0.25
Freshness	0.10	0.10	0.10	0.10
Consistency	0.15	0.10	0.10	0.10
Entity match	0.10	0.10	0.10	0.10
Population match	0.05	0.05	0.05	0.03
Anti-poisoning	0.10	0.05	0.03	0.01
Anti-injection	0.05	0.05	0.02	0.01
Threshold τ_r	0.30	0.45	0.60	0.75

and as the anti-poisoning and anti-injection factors, so a high trust score cannot buy back a security flag. The trust score filters chunks; it does not reorder them, so eligible evidence keeps its fused rank.

C. Claim verification and release

Eligible chunks are numbered and passed to the generator as data, and each claim must cite them with `[Source N]` markers. Each generated claim c is then checked four ways. A pinned NLI model (PubMedBERT fine-tuned on MNLI and MedNLI, at a fixed revision [22], [23]) judges each clause against its cited chunk, so one unsupported clause makes the claim unsupported; an aggregate check considers all retrieved chunks. A pair-identity check compares the drug pair, relation and direction with an approved relationship record. Citation checks resolve each marker to an eligible chunk and check alignment and claim strength. The medication identity gate requires every medication or class that the claim puts in a drug role to be named word for word by a cited source (text or verified title), resolving anaphors and rejecting unresolved or invented names; a separate guard rejects medications the user did not confirm. The release predicate is

$$\begin{aligned}\mathrm{Release}(c) = [\, s(c) = \textsc{supported} \,] \wedge [\, g(c) = \textsc{pass} \,] \\ \wedge\, [\, m(c) = \varnothing \,],\end{aligned} \tag{5}$$

where s is the final support state, g the gate decision and m the guard message. The predicate reads the gate and guard records directly, so no later stage can restore a rejected claim, and a claim without a gate record fails. Withheld claims produce a structured abstention with the stage and reason.

V. ALGORITHMS

Algorithms 1 and 2 give the two stages as implemented. Admission applies every exclusion before the trust score, so a chunk that fails one is never scored, and an empty evidence pack ends the request with an abstention before any model is called. Verification runs the verifier and the identity gate on every claim, and the strength, citation-alignment and guard checks on every claim the verifier passes, and records each result even when an earlier check has already failed; release requires all of them (5). Because the results are recorded separately, the release decision under any subset of the checks can be recomputed from stored runs (Section VIII-D).

Algorithm 1 Evidence admission (pre-generation)

Require: query Q, confirmed medications D; tier thresholds τ
Ensure: numbered evidence pack $\mathcal{P}$ or abstention
screen and sanitise Q; resolve D in the pinned lexicon
$\mathcal{E} \leftarrow$ RRF of BM25, dense and graph rankings (2); drop poisoning score > 0.4
$r \leftarrow$ risk tier of Q; $\mathcal{P} \leftarrow \emptyset$
for each $e_i \in \mathcal{E}$ in fused order **do**
  if provenance blocked **or** hash mismatch **or** carries instructions **then continue**
  end if
  if e_i names no medication in D **or** identity unverified **then continue**
  end if
  if text already in $\mathcal{P}$ **then continue**
  end if
  $T_i \leftarrow$ (3) with missing factors set to 0
  if $T_i \geq \tau_r$ **and** $A_i \geq 0.3$ **then** $\mathcal{P} \leftarrow \mathcal{P} \cup \{e_i\}$
  end if
end for
if $\mathcal{P} = \emptyset$ **then return** abstention (no eligible evidence)
end if
return $\mathcal{P}$ numbered as [Source 1..N]

Algorithm 2 Claim verification and release (post-generation)

Require: answer y generated from $\mathcal{P}$; confirmed medications D
Ensure: released claims and abstentions with reasons
for each claim c in y **do**
  $s(c) \leftarrow$ clause-level NLI on cited chunks, aggregate check, pair identity, citation and strength checks
  $g(c) \leftarrow$ identity gate on drug-role names vs. cited text and titles
  $m(c) \leftarrow$ guard: medications named but not in D
  if (5) holds **then** release c
  else withhold c and record the first failing stage
  end if
end for

VI. DATA AND EVIDENCE MANAGEMENT

Every chunk carries a provenance record: source type, document identifier, URL, retrieval time, content hash and, where established, a source-identity record with title and version. The evaluated local corpus has 17 items. All 17 content hashes match their text, but only the three DailyMed label sections (lisinopril sections 5.5 and 7.1, warfarin sodium section 7.3) have a verified source identity; the other 14 chunks have none and 11 carry placeholder URLs, and no relationship record has owner approval. A matching hash shows that stored text is unchanged; it does not show that the text came from the claimed source. DailyMed shows each product's in-use labeling, which may differ from the latest FDA-approved labeling, so set identifier, version, section and retrieval date

are kept with each item.

VII. Evaluation Method

Apart from the pilot in Section VIII-H, all runs used an isolated copy of the frozen candidate, offline, with the pinned NLI model and no provider call. Generation was scripted: each test statement is fed through the real release path as if a model had written it, so the checks are measured, not a language model. A statement can stop in four places: before the identity gate (no eligible evidence), at the gate, after it, or released; counting an upstream abstention as a gate rejection would credit the gate for work it did not do, so we report each separately. Seven constructed statement sets (V2 to V8, 731 scored rows; row counts in Table II) were written during development, each after the previous candidate had been tested, so they are development-informed regression sets; V8 was written by the author of C2h and is author-known. Each row is labelled PASS (every medication or class in a drug role is named by the cited source) or REJECT, with the expected failing entity. All labels are AI-written and unreviewed; blind review packets with opaque identifiers are prepared.

For each set we report the gate miss rate (REJECT rows the gate passed, over REJECT rows that reached it), unsupported releases (REJECT rows released), the identity false-withhold rate (PASS rows the gate rejected, over PASS rows) and coverage (PASS rows released). Rates carry 95% intervals: Clopper and Pearson's exact interval [25] when the count is 0 and Wilson's interval [24] otherwise. Paired comparisons on the same rows use the exact McNemar test [26]. Sets are never pooled into one rate. The acceptance criterion for identity false withholds, fixed before any new confirmation set, is a point estimate of at most 5% with the Wilson interval reported.

VIII. Results

A. Reproducibility

A clean offline rerun of the frozen candidate produced 15 of 15 result files byte-identical to the stored reference, and the backend suite gave 2,252 passed, 0 failed and 4 skipped (provider credentials absent) in normal and network-blocked modes (run `C2H_REPRO_01`).

B. Identity gate

Table II and Fig. 2a give the per-set results. No REJECT row was released in any set. Among REJECT rows that reached the gate, the gate passed one (V6-IND-03, a row whose label is a recorded error). One REJECT row (V7-SRC-02) never reached the gate: its cited section names neither requested medication, so the pipeline abstained before generation. On V8 the two false withholds came from the ordinary words "nudge" and "oozing", which the gate's word list did not contain (Fig. 3b).

C. Where valid statements stop

Coverage is low (Fig. 2b). Across V2 to V8, 206 valid statements were released, 5 were rejected by the identity gate and 204 passed the gate and were withheld later (run `STOP_DIAGNOSTICS_C2H_01`). The first stage to withhold them was NLI in 129 (entailment below 0.10 in 101), pair identity in 52, the verifier's aggregate check in 18 and the medication guard in 5. The pair-identity stops follow from missing or unapproved relationship records (34 claims whose canonical identity was not resolved, 18 whose source identity was unavailable), so they are correct under the current evidence. Whether the NLI and aggregate stops are right cannot be decided without claim-support labels. The five guard stops are a defect: the guard reported the ordinary words "Perform" (three times), "Align" and "sodium" as unconfirmed medications although the gate had passed them. "Perform" and "Align" are RxNorm brand names opening an instruction ("Perform potassium monitoring periodically"), and "sodium" occurs in "a low-sodium diet".

TABLE II
Identity gate per development set (C2h; runs `STOP_POINTS_SUPPLEMENTARY_01`, `OFFLINE_METRICS_01`)

Set	Gate miss (95% CI, %)	Not reached	False withhold (95% CI, %)	Coverage
V2	0/40 (0, 8.8)	0	0/79 (0, 4.6)	45/79
V3	0/36 (0, 9.7)	0	2/65 (0.9, 10.5)	40/65
V4	0/28 (0, 12.3)	0	0/44 (0, 8.0)	31/44
V5	0/36 (0, 9.7)	0	0/48 (0, 7.4)	26/48
V6	1/61 (0.3, 8.7)	0	1/65 (0.3, 8.2)	34/65
V7	0/60 (0, 6.0)	1	0/55 (0, 6.5)	13/55
V8	0/54 (0, 6.6)	0	2/59 (0.9, 11.5)	17/59

Unsupported releases: 0 in every set. Labels AI-written; sets development-informed.

D. What each check contributes

Because every check result is recorded, we recomputed the release decision of every development row under subsets of the checks (Table III and Fig. 3a; run `CHECK_COUNTERFACTUAL_01`). Recomputing the full rule reproduced the recorded outcome on all 731 rows. A conventional pipeline (1) fed the same scripted statements shows all of them, including every REJECT row. With every check except the identity gate, 3 to 19 identity-unsupported statements per set would have been released (75 rows in the seven sets). In these rows the NLI model found the claim entailed by its cited chunk, but the chunk does not name a medication or class that the claim asserts: invented names ("Duskwell can raise the bleeding hazard for warfarin users") and classes that the warfarin or lisinopril section does not name ("SSRIs may call for closer INR follow-up in warfarin users"). The identity gate alone released one REJECT row (V6-IND-03, the recorded label error) and withheld no valid row apart from its own false withholds, but it checks no semantic support; the full rule released 13 to 45 valid rows per set. The sets were written to exercise the gate, so these counts show what the other checks miss on such statements, not how often such statements occur in real answers.

E. Candidate comparison and remediation

On the same 113 V8 statements (Fig. 2c), the previous candidate's code (C2g) had 4 gate misses, 1 unsupported

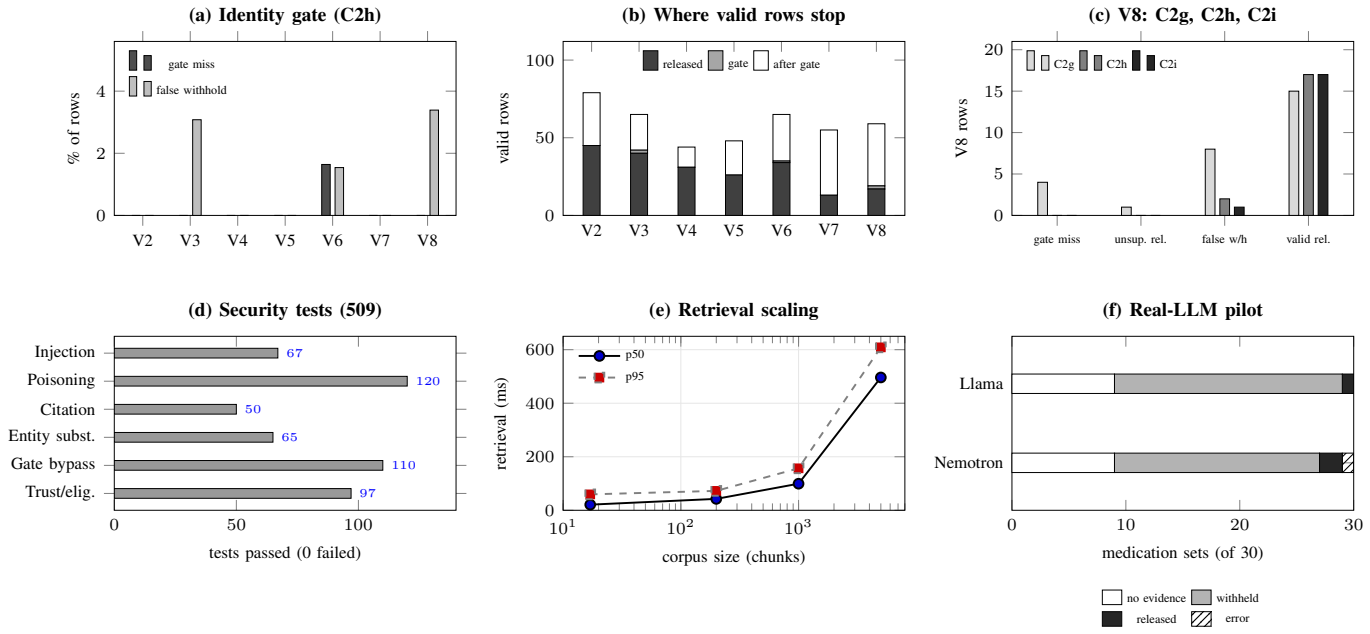


Fig. 2. Results at a glance. (a) Gate misses and identity false withholds per development set (C2h; exact intervals in Table II). (b) Where valid statements stop (`OFFLINE_METRICS_01`). (c) Three code versions on the same V8 statements (`V8_PAIRED_C2G_C2H_01`, `C2H_VS_C2I_02`; V8 is author-known). (d) Scoped security tests by threat group, all passing on C2h, C2i and C2j. (e) Hybrid retrieval latency on synthetic corpora (`PF01_SCALING_01`). (f) Terminal outcome per medication set in the real-LLM pilot (`LIVE_TEXT_01`, `_03`).

TABLE III
ROWS RELEASED UNDER SUBSETS OF THE CHECKS, RECOMPUTED FROM RECORDED RESULTS (C2H; RUN `CHECK_COUNTERFACTUAL_01`)

	REJECT rows released				PASS rows released		
Set	none	gate only	all but gate	all	of	gate only	all
V2	40/40	0	14	0	79	79	45
V3	36/36	0	14	0	65	63	40
V4	28/28	0	5	0	44	44	31
V5	36/36	0	3	0	48	48	26
V6	61/61	1	19	0	65	64	34
V7	61/61	0	10	0	55	55	13
V8	54/54	0	10	0	59	57	17

"None": no eligibility stage and no check, as in (1); the other columns keep the eligibility stage, which stopped V7-SRC-02. Scripted generation; labels AI-written; sets development-informed.

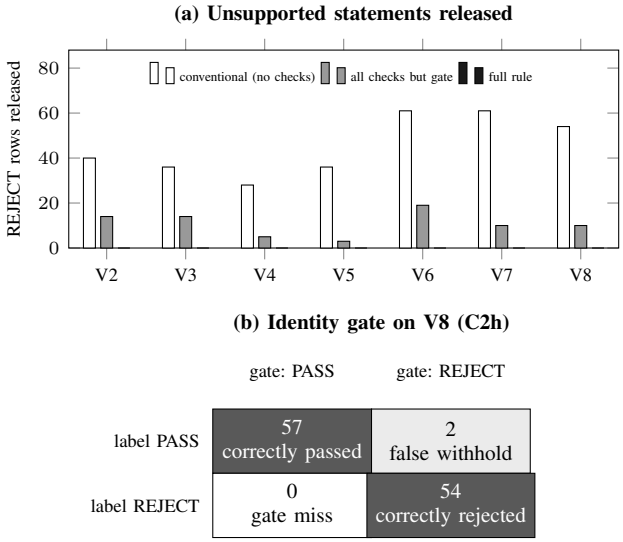


Fig. 3. (a) Unsupported (REJECT) statements released per development set by a conventional pipeline, by every check except the identity gate, and by the full release rule, recomputed from recorded results (C2h; `CHECK_COUNTERFACTUAL_01`; Table III). (b) Confusion matrix of the identity gate on the 113 V8 statements (Table II); the two false withholds are the words "nudge" and "oozing". Labels are AI-written and V8 is author-known.

release, 8 false withholds and 15 valid releases; C2h had 0, 0, 2 and 17. C2h fixed 10 decisions and broke none (exact McNemar $p = 0.002$; runs `V8_PAIRED_C2G_C2H_01`, `V8_RISK_COVERAGE_01`); since V8 is author-known, this is exploratory.

We then built a remediation candidate (C2i) from C2h to fix the two defects found above without loosening any check. First, an ordinary verb is now known when its listed past or present participle is (the word list held "nudged" but not "nudge"); plural forms are not used, so a name whose plural happens to be listed is not made known. Second, the guard skips an unconfirmed finding only in two forms: an RxNorm brand name that is an ordinary English word, opens a sentence and is not followed by a modal or auxiliary ("Perform potassium monitoring", but not "Perform may raise …"), and an ingredient inside a "low-", "high-" or "reduced-" diet compound. A brand used anywhere else, a brand that is not an ordinary word ("Advil"), an everyday-word ingredient such as lithium, and every confirmed or unresolved name are still reported; 52 new unit tests pin these cases. Across

all 731 development rows six decisions changed, none for the worse (run C2H_VS_C2I_02): the five guard rows (H-CWD-04, H-CWD-08, V3-CWD-03, V5-CW-04, V6-NPH-04) are now released, and the "nudge" statement (V8-SRC-08) passes the gate but is still withheld by pair identity. No unsupported statement was released, the 42-example NLI replay was unchanged, the 509 security tests passed and the backend suite gave 2,304 passed, 0 failed and 4 skipped with the network blocked (run C2I_REGRESSION_02). "Oozing" remains unfixed because no vocabulary we could audit offline contains any form of "ooze". V2 to V8 informed these changes, so these are development-set results.

F. Security regression

We grouped 509 existing tests by threat and ran them with the network blocked (Fig. 2d): prompt injection 67, poisoning and provenance 120, citation manipulation 50, entity substitution 65, gate bypass and fallback 110, trust and eligibility 97. All passed on C2h, on C2i and on C2j (runs SECURITY_REGRESSION_01, C2J_PORT_CANCELLATION_01). The tests check specified behaviours with inputs written by the developers; they are not an attack-success rate, and a 110-case security set stays unrun until people review its labels.

G. Performance

On one CPU machine with a warm file cache and the 17-item corpus, median latency was 7.9 ms for retrieval ($n = 63$), 1.4 ms for the identity gate per statement ($n = 635$) and 164.6 ms for the full offline release path ($n = 339$; 95th percentile 336.7 ms); peak working set was 837 MB (run PF01_OFFLINE_LATENCY_02). These figures exclude any language-model call. On C2j (PF01_C2J_SUPPLEMENT_01) one NLI pair took 254.9 ms (median; p95 478.9 ms), local OCR 1.05 s per image, grounding under 2 ms and name normalisation 0.03 ms.

To see how retrieval scales, we built the unmodified hybrid retriever over synthetic, fictional corpora of 17, 200, 1,000 and 5,000 chunks and timed the same 30 queries after 5 warm-up queries (run PF01_SCALING_01; Fig. 2e). Median latency grew from 21.0 ms to 42.6, 99.3 and 496.2 ms, roughly linearly with corpus size, because the dense channel scores every chunk by brute-force cosine similarity; building the index (tokenising and embedding) took 1.2 s, 7.7 s, 39.1 s and 183.7 s, and peak working set reached 980 MB at 5,000 chunks. This harness encodes each query with the sentence-transformer directly, so its 17-chunk figure is not comparable with the 7.9 ms above. A production corpus would need an approximate nearest-neighbour index; the synthetic text says nothing about retrieval quality.

H. Pilot with real language models

Everything above used scripted generation. To see how the release path handles text a model actually writes, we ran a small authorised pilot on C2i with two hosted models, Llama 3.3 70B Instruct on Cloudflare Workers AI and Nemotron 3 Ultra on NVIDIA's API. Each received the same 30 synthetic medication sets once: 15 with warfarin, 5 with lisinopril and 10 control sets with no evidence in the corpus. Only synthetic inputs and the three public DailyMed sections were sent, and the NLI model ran locally. NVIDIA's Nemotron 3 Super 120B and Llama 3.1 Nemotron 70B were not available to the project's account (LIVE_TEXT_01, _02).

TABLE IV
REAL-MODEL PILOT ON 30 SYNTHETIC MEDICATION SETS, ONE RUN PER MODEL (C2I; RUNS LIVE_TEXT_01, LIVE_TEXT_03, SCORED IN LIVE_SUMMARY_01)

	Llama 3.3 70B (Cloudflare)	Nemotron 3 Ultra (NVIDIA)
Stopped before generation	9	9
Withheld after generation	20	18
Answer released	1	2
Provider error	0	1 (timeout)
Control sets released	0/10	0/10
Claims generated	141	171
Claims passing every per-claim check	51	82
Claims released	10	2
Failed claims without a relationship record*	38	53
Failed claims with a different relationship*	4	9
Failed claims rejected by the identity gate*	39	14
Time per generated case, median / p95 (s)	9.9 / 30.6	20.2 / 87.8

*As recorded in the abstention reason; one claim can carry several reasons.

Table IV and Fig. 2f give the outcomes. Nine control sets per model stopped before generation; the tenth (sertraline with omeprazole) reached both models because the warfarin section names sertraline, and both answers were withheld for claiming that the two drugs do not interact. The release rule is all or nothing, so one failing claim withholds the whole answer. Llama 3.3 had one answer released (10 claims, lisinopril with metformin) and Nemotron two (one claim each: warfarin with ibuprofen, requested as Coumadin and Advil, and warfarin with atorvastatin). Most failed claims had no approved relationship record, as in the scripted runs; the identity gate rejected 39 Llama claims and 14 Nemotron claims. Each of the 12 released claims restates its cited DailyMed section, and the lisinopril answer says nothing about metformin, for which the corpus has no evidence; we checked this ourselves, not independently.

For image reading we drew 20 synthetic prescriptions, five each plain, with brand and generic names, rotated and noisy, and sent each once (Table V). The application's earlier default, Llama 3.2 11B Vision, had been retired (HTTP 410 on all attempts), and Llama 3.2 90B Vision timed out. Nemotron 3.5 Super VL answered 14 images and read 35 of their 36 names with no unrelated name; 13 images were exact. Its one error matters here: on a rotated image it returned "metoprolol" for metoprolol tartrate, dropping the salt. The other six replies were empty or not JSON and failed closed. Median time was 7.1 s per image (95th percentile 32.7 s). We made this model the default in C2i; the full regression then gave the same V2 to V8 decisions, 509 of 509 security tests

TABLE V
PRESCRIPTION-IMAGE EXTRACTION, 20 SYNTHETIC IMAGES, ONE ATTEMPT EACH (NVIDIA API; RUN LIVE_VISION_04 UNLESS STATED)

Model and image type	Images	Answered	Exact	Failed closed
Llama 3.2 11B Vision (_01)	20	0	–	20 (HTTP 410)
Llama 3.2 90B Vision (_02)	3	0	–	3 (timeout)
Nemotron 3.5 Super VL				
plain	5	3	3	2
brand and generic	5	4	4	1
rotated	5	2	1	3
noisy	5	5	5	0
all	20	14	13	6

Names on answered images: 35 of 36 read; 1 returned without its salt.

TABLE VI
C2J CHANGES, EACH FOLLOWED BY A FULL REGRESSION: 0 FAILED, 4 SKIPPED IN BOTH MODES; 509/509 SECURITY TESTS (BASELINE C2I: 2,307 PASSED, C2J_BASELINE_REPRO_02)

Change	Run C2J_PORT_	Passed
Single-worker startup guard	SINGLE_WORKER_02	2,316
Provider error bodies not logged	ERROR_BODIES_01	2,322
OCR-first reading, name grounding	OCR_FIRST_02	2,377
Cancellation and request lifecycle	CANCELLATION_01	2,415

and 2,307 backend tests passed, 0 failed, 4 skipped, network blocked and in normal mode (C2I_REGRESSION_03, C2J_BASELINE_REPRO_02).

These pilots are small and written by us, so they show behaviour, not answer quality: on real model output the release path withheld most answers and released none for the controls.

I. Hardening of the remediation candidate (C2j)

The final candidate, C2j, adds four changes to C2i from a parallel line of the same application (Table VI); none touches retrieval, trust scoring or verification. The server refuses to start with more than one worker; provider error bodies, which could carry prescription text or the account identifier, are no longer logged; images are read by local OCR first and vision-model names are grounded in that text; and a cancel endpoint, a disconnect handler and a 120 s deadline end abandoned analyses and their OCR process. After each change the full regression kept all 731 V2 to V8 decisions and the 42-example replay byte-identical, so the development results above hold for C2j, and a comparison of function bodies found the release rule unchanged. The frontend passed 148 of 148 tests (C2J_FRONTEND_02) and 154 tests with the real NLI model loaded passed (C2J_REAL_NLI_SUITE_02).

An offline replay of the vision pilot's recorded answers (C2J_FINAL_METRICS_01) found OCR text on all 20 images (median 143 characters), so all would still be sent; grounding kept all 35 correct names and also "metoprolol" for metoprolol tartrate, whose missing salt only the user's confirmation can catch.

IX. IMPLEMENTATION STATUS

Table VII records each component of the final candidate and the evidence behind it: "tests" means the passing backend suite, "dev sets" means measured on V2 to V8, and "pilot" or "replay" the small real-model runs. No component has yet been validated on independent data with independent reviewers.

TABLE VII
COMPONENTS OF THE FINAL CANDIDATE (C2J) AND THE EVIDENCE FOR EACH

Component	Evidence
Query screening; exclusion checks (hash, injection, identity)	tests; 509 security tests
Medication normalisation (RxNorm lexicon)	tests; latency
Hybrid retrieval and RRF; trust scoring and tiers	tests; latency
Clause-level NLI; pair identity and citation checks	tests; dev sets (stop points)
Medication identity gate; release rule and abstention	tests; dev sets; LLM pilot
C2i vocabulary and guard fixes	tests; dev sets (6 of 731 changed)
Prescription image reading (vision model)	tests; vision pilot (20 images)
OCR-first reading and name grounding (C2j)	tests; offline replay
Cancellation, single-worker guard, log redaction (C2j)	tests
Evidence-source adapters (not in the live path)	tests (snapshot mode)

X. LIMITATIONS AND FUTURE WORK

The results describe software behaviour on constructed data. No qualified person has reviewed any label, and V2 to V8 were written during development, so the identity results are preliminary and do not show generalisation; an independently written, blind-reviewed confirmation set is needed. Generation was scripted except in the pilot, which used one run per model on 30 cases and 20 images that we wrote ourselves, so real model output is barely sampled. The local corpus is not a gold corpus (14 of 17 chunks without verified identity), so retrieval quality, claim-level support and citation support remain unmeasured, and so does the comparison with semantic-similarity RAG that research question 1 asks for. The security tests do not measure robustness to unseen attacks, and latency was measured on one machine. The C2j changes were checked by tests and an offline replay only, not with users or real prescriptions. Table VIII summarises what the evidence supports. Each open question needs something software alone cannot supply: retrieval quality and claim support need a verified corpus and independent relevance annotators, generalisation needs a confirmation set written and reviewed by others, the comparison with semantic RAG needs both, and clinical correctness needs qualified clinical reviewers. A learned adversarial detector would need labelled training data.

The next evaluation is designed but not run: a frozen corpus of at least 200 verified chunks from 30 or more documents; dense, BM25, hybrid and hybrid-with-trust retrieval compared on the same queries; two blind annotators grading relevance with adjudication and weighted κ; and P@K, R@K, MRR

TABLE VIII
WHAT THE EVIDENCE SUPPORTS

Claim	Status
Offline reproducibility of the evaluated candidates	supported
Scoped security regression (509 tests)	supported
Engineering hardening of C2j (tests, run records)	supported
Offline latency (one machine)	supported
Corpus provenance (descriptive)	supported
Stop-point diagnosis of withheld statements	supported
Identity gate, C2i fixes, contribution of each check	preliminary
Coverage and abstention behaviour	partial
Release path on real LLM output; image reading	pilot

Open questions, not yet measurable: retrieval quality, claim and citation support, independent generalisation, comparison with semantic RAG, clinical correctness (Section X).

and nDCG with bootstrap intervals. The metric code already passes 15 hand-calculated tests.

XI. CONCLUSION

The prototype separates medication identity, provenance and semantic support, combines them in a release rule that no later stage can override, and records where each statement stops. On constructed development data its identity gate released no unsupported statement, at the price of low coverage that mostly comes from checks after the gate, and recomputing the release decision from the recorded checks showed that the gate catches misattributed statements that semantic entailment alone accepts. Whether trust-aware retrieval beats semantic retrieval is still open and needs the verified corpus, independent confirmation set and human relevance judgments described above. The system is a research prototype and is not intended for clinical use.

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REFERENCES

[1] P. Lewis *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.

[2] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, and M. Wang, "Retrieval-augmented generation for large language models: A survey," arXiv:2312.10997, 2023.

[3] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi, "Self-RAG: Learning to retrieve, generate, and critique through self-reflection," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024. arXiv:2310.11511.

[4] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, "Benchmarking retrieval-augmented generation for medicine," in *Findings Assoc. Comput. Linguistics: ACL 2024*, 2024, pp. 6233–6251, doi: 10.18653/v1/2024.findings-acl.372.

[5] R. P. Radha Krishnan *et al.*, "Evaluating the capability of ChatGPT in predicting drug-drug interactions: Real-world evidence using hospitalized patient data," *Br. J. Clin. Pharmacol.*, vol. 90, no. 12, pp. 3361–3366, 2024, doi: 10.1111/bcp.16275.

[6] J. Sicard *et al.*, "Can large language models detect drug-drug interactions leading to adverse drug reactions?" *Ther. Adv. Drug Saf.*, vol. 16, Art. no. 20420986251339358, 2025, doi: 10.1177/20420986251339358.

[7] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection," in *Proc. 16th ACM Workshop Artif. Intell. Secur. (AISec)*, 2023, pp. 79–90, doi: 10.1145/3605764.3623985.

[8] W. Zou, R. Geng, B. Wang, and J. Jia, "PoisonedRAG: Knowledge corruption attacks to retrieval-augmented generation of large language models," in *Proc. 34th USENIX Secur. Symp.*, 2025, pp. 3827–3844.

[9] J. B. Hakim *et al.*, "The need for guardrails with large language models in pharmacovigilance and other medical safety critical settings," *Sci. Rep.*, vol. 15, Art. no. 27886, 2025, doi: 10.1038/s41598-025-09138-0.

[10] J. C. L. Ong *et al.*, "Large language model as clinical decision support system augments medication safety in 16 clinical specialties," *Cell Rep. Med.*, vol. 6, no. 10, Art. no. 102323, 2025, doi: 10.1016/j.xcrm.2025.102323.

[11] Y. Sun, W. Wang, W. Cheng, R. Li, X. Liu, Q. Pei, J. Tang, and G. Yang, "Retrieval-augmented generation for medication safety: A case study using drug package inserts," *Int. J. Med. Inform.*, vol. 220, Art. no. 106599, 2026, doi: 10.1016/j.ijmedinf.2026.106599.

[12] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto, and P. Fung, "Survey of hallucination in natural language generation," *ACM Comput. Surv.*, vol. 55, no. 12, pp. 1–38, 2023, doi: 10.1145/3571730.

[13] C. Niu *et al.*, "RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (ACL)*, 2024, pp. 10862–10878, doi: 10.18653/v1/2024.acl-long.585.

[14] S. Min *et al.*, "FActScore: Fine-grained atomic evaluation of factual precision in long form text generation," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2023, pp. 12076–12100, doi: 10.18653/v1/2023.emnlp-main.741.

[15] OWASP Foundation, "OWASP Top 10 for large language model applications," 2025. [Online]. Available: https://genai.owasp.org/llm-top-10/

[16] S. J. Nelson, K. Zeng, J. Kilbourne, T. Powell, and R. Moore, "Normalized names for clinical drugs: RxNorm at 6 years," *J. Am. Med. Inform. Assoc.*, vol. 18, no. 4, pp. 441–448, 2011, doi: 10.1136/amiajnl-2011-000116.

[17] V. Karpukhin, B. Oğuz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W.-t. Yih, "Dense passage retrieval for open-domain question answering," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2020, pp. 6769–6781, doi: 10.18653/v1/2020.emnlp-main.550.

[18] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP-IJCNLP)*, 2019, pp. 3980–3990, doi: 10.18653/v1/D19-1410.

[19] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," *Found. Trends Inf. Retr.*, vol. 3, no. 4, pp. 333–389, 2009, doi: 10.1561/1500000019.

[20] G. V. Cormack, C. L. A. Clarke, and S. Buettcher, "Reciprocal rank fusion outperforms Condorcet and individual rank learning methods," in *Proc. 32nd Int. ACM SIGIR Conf. Res. Develop. Inf. Retr.*, 2009, pp. 758–759, doi: 10.1145/1571941.1572114.

[21] S. R. Bowman, G. Angeli, C. Potts, and C. D. Manning, "A large annotated corpus for learning natural language inference," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2015, pp. 632–642, doi: 10.18653/v1/D15-1075.

[22] Y. Gu *et al.*, "Domain-specific language model pretraining for biomedical natural language processing," *ACM Trans. Comput. Healthc.*, vol. 3, no. 1, pp. 1–23, 2021, doi: 10.1145/3458754.

[23] A. Romanov and C. Shivade, "Lessons from natural language inference in the clinical domain," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2018, pp. 1586–1596, doi: 10.18653/v1/D18-1187.

[24] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *J. Am. Stat. Assoc.*, vol. 22, no. 158, pp. 209–212, 1927, doi: 10.1080/01621459.1927.10502953.

[25] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934, doi: 10.1093/biomet/26.4.404.

[26] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947, doi: 10.1007/BF02295996.